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Assessment of the electron donor properties of substituted phenanthroline ligands in molybdenum carbonyl complexes using molecular electrostatic potentials†

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The relevance of substituent effects in the design and development of novel molecules has considerable interest in many industrial and synthetic reactions in organic chemistry. Molecular electrostatic potential (MESP) parameters are found to be excellent descriptors for characterizing the substituent and ligand effects in chemistry. Here we use MESP parameters derived for substituted benzenes as a measure of the substituent effect to assess the coordination ability of substituted 1,10-phenanthrolines with Mo(CO)₄. The MESP approach used to measure the substituent effect provides an easy way to predict ligand effects before any theoretical calculations or experimental verification and it is considered to be the most accessible guideline for the quantification of the electron donating power of aromatic N-heterocyclic ligands.

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Introduction

Substitution chemistry has grown to become an attention-grabbing subject in the field of organic synthesis while quantification of substituent effects is considered as very crucial in deducing the structure–activity relationships.^{1–3} Generally, Hammett constants^{4–7} are used to convey the ability of a substituent to attract or lose electrons. Hammett constants are derived based on the ionization of substituted benzoic acids. Quantum mechanically derived substituent parameters are commonly used now, instead of the experimentally observed and classical mechanics based parameters.^{8–11} Among such parameters, the molecular electrostatic potential (MESP) has materialized as a remarkable descriptor of the electronic properties of molecules.^{12–16} Our previous studies have shown that MESP parameters behave very similarly to Hammett constants in describing the substituent effects in chemistry.^{17–20} In a recent study, 381 substituents were classified into different categories, such as very strong, strong, weak, and very weak based on their electron donating/withdrawing ability measured using the change in MESP value at the para carbon (ΔV_C) of a substituted benzene with respect to benzene.¹⁸ The entire database of V_C deduced for benzene is found to be transferable to π -conjugated systems such as butadiene, butadiyne,

pyridine and pyrene. The aim of the present study is to show that the ΔV_C database is useful for finding structure–activity relationships in molecular complexes. Here we try to deduce such a relationship for selected substituted 1,10-phenanthrolines (phen*) which is a key class of bidentate chelating ligands in inorganic chemistry.^{21,22} The ligand is said to be highly versatile due to its planar structure, aromaticity and strong chelating ability.²³ Very recently, the donor–acceptor properties of a large set of substituted 1,10-phenanthrolines²⁴ have been experimentally and theoretically determined by Brenna *et al.*, and they reported a significant amount of data in terms of energetic contributions to describe the bonding between molybdenum tetra carbonyl, Mo(CO)₄, and 1,10-phenanthrolines. The present study uses the data reported by Brenna *et al.* to verify the MESP based approach to assess substituent effects in 1,10-phenanthrolines. To be aware of the coordination behaviour of metal complexes, it is indispensable to gain knowledge about the metal–ligand interactions.^{25,26} The MESP minimum ($V_{\min}$) at the lone pair bearing nitrogen atom of the phen* ligand and the MESP at the nitrogen nucleus (V_N) have been used here to predict the interactive behaviour of phen* with Mo(CO)₄ to form [Mo(CO)₄(phen*)]. In our previous works, we have used such MESP parameters to quantify the electronic behaviour of N-heterocyclic carbene ligands and phosphine ligands.^{27,28}

Computational details

All the computational calculations have been performed with density functional theory (DFT) using the Gaussian 09 suite of programmes.²⁹ The 1,10-phenanthroline ligands are optimized

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at the B3LYP level^{30,31} with the 6-311+G(d,p) basis set.³² For optimizing the molybdenum complexes, the QZVP basis set is used for Mo. All the structures are verified as energy minima by frequency calculations. The MESP, $V(\mathbf{r})$,^{33–35} in Hartree units is defined in eqn (1)

$$V(\mathbf{r}) = \sum_A^N \frac{Z_A}{|\mathbf{r} - \mathbf{R}_A|} - \int \frac{\rho(\mathbf{r}') d^3\mathbf{r}'}{|\mathbf{r} - \mathbf{r}'|} \quad (1)$$

where Z_A is the nuclear charge, located at $\mathbf{R}_A$, and $\rho(\mathbf{r}')$ is the electron density. The two terms in eqn (1) stand for the nuclear and electronic contributions respectively. Pioneering works of Gadre *et al.* have shown the usefulness of topography mapping of molecular electrostatic potential^{36,37} for the fine tuning of molecular reactivities, electronic structure and intermolecular interactions. The DAMQT package is used for the topographical analysis of MESP.³⁸ The descriptors obtained from the molecular electrostatic potential (MESP) through the topographical analysis are advantageous for the design of new QSAR models.^{39,40}

Results and discussion

(a) Design strategies

The thirteen 1,10-phenanthroline derivatives (phen* ligands) selected for this study used for forming $[\text{Mo}(\text{CO})_4(\text{phen}^*)]$ complexes are depicted in Fig. 1. Synthesis of all these ligands has recently been reported by Brenna *et al.* The phen* ligands can be distinguished on the basis of the nature of substituents and their positions on the phenanthroline framework. All the ligands are named according to the naming scheme given by Brenna *et al.* We use these systems as a test set to analyze the MESP features to find correlation with experimental data. The ΔV_C parameter reported in the previous study, derived from substituted benzenes will also be tested for finding the transferability of the MESP property to describe the substituent effect in 1,10-phenanthrolines and their Mo complexes. The experimental IR stretching frequency A_{Iax} of the CO groups in $[\text{Mo}(\text{CO})_4(\text{phen}^*)]$, reported by Brenna *et al.*,²⁴ is selected as the experimental electronic parameter to find correlation with MESP features. Further, interaction energy (E_{int}) due to the

coordination of 1,10-phenanthrolines to $\text{Mo}(\text{CO})_4$ is calculated and correlated with the substituent effect parameters (Fig. 4). The correlations found in the test set will be applied to a test set containing 108 phenanthroline derivatives to make predictions on their E_{int} . Moreover, the predictions will be verified by calculating E_{int} of many representative cases. Finally, a randomization procedure on the substitution pattern will be adopted to predict the substituent effect of 5000 multiply substituted 1,10-phenanthrolines.

(b) MESP analysis of substituted 1,10-phenanthroline ligands

The MESP parameters ΔV_N and ΔV_{min} used for interpreting the interaction energy of $[\text{Mo}(\text{CO})_4(\text{phen}^*)]$ complexes are defined in eqn (2) and (3), respectively. ΔV_N accounts for the difference between the V_N of phen* and phen ligands while ΔV_{min} describes the difference between the corresponding V_{min} values. The MESP minimum for the carbonyl oxygen ($V_{\text{min-CO}}$) in the complex $[\text{Mo}(\text{CO})_4(\text{phen}^*)]$ is also analyzed. The interaction energy (E_{int}) of the $[\text{Mo}(\text{CO})_4(\text{phen}^*)]$ complexes is calculated by subtracting the energy of $[\text{Mo}(\text{CO})_4(\text{phen}^*)]$ from the sum of the energies of the fragments $\text{Mo}(\text{CO})_4$ and phen* (eqn (3)).

$$\Delta V_N = V_N(\text{phen}^*) - V_N(\text{phen}) \quad (2)$$

$$\Delta V_{\text{min}} = V_{\text{min}}(\text{phen}^*) - V_{\text{min}}(\text{phen}) \quad (3)$$

$$E_{\text{int}} = E[\text{Mo}(\text{CO})_4(\text{phen}^*)] - \{E[\text{Mo}(\text{CO})_4] + E[\text{phen}^*]\} \quad (4)$$

The MESP features of the phen* ligands are presented in Fig. 2 using suitable isosurfaces. The $6.0 \text{ kcal mol}^{-1}$ valued isosurface (Fig. 2a) shows the π -electron region of the aromatic rings and the lone pair site of the nitrogen centers. Also localization of a double bond for the central arene ring is visible in the MESP while a lower MESP value shows electron rich nitrogen centers and the lowest point (V_{min}) indicates the lone pair centers of the nitrogen atoms (Fig. 2b). V_{min} represents a (3, +3) critical point (CP) in the

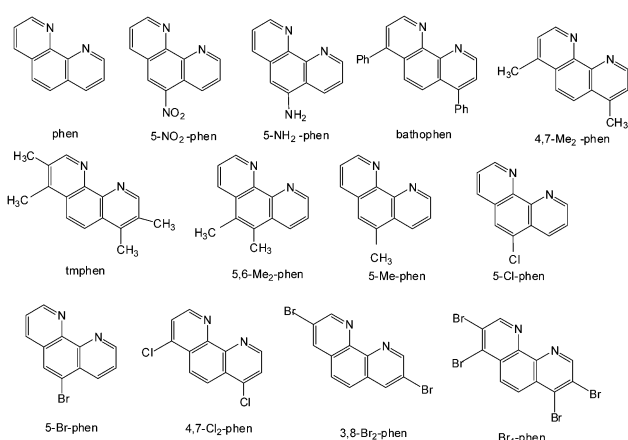


Fig. 1 Experimentally known substituted 1,10-phenanthroline ligands.

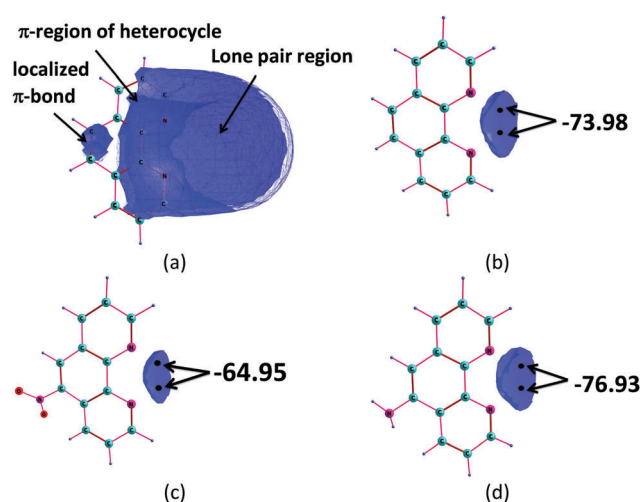


Fig. 2 MESP plots of the phen ligand are shown with (a) 6 kcal mol^{-1} isosurface of the phen ligand, (b) MESP plots of the phen ligand with 44 kcal mol^{-1} isosurface and V_{min} points, (c) MESP plots of the NO_2 -phen ligand with 44 kcal mol^{-1} isosurface and V_{min} points and (d) MESP distribution of the NH_2 -phen ligand with 44 kcal mol^{-1} isosurface and V_{min} points.

MESP topography. The three positive eigenvalues of $V_{\min}$ indicate how sharply the function is increasing in all the three directions at the CP while the corresponding eigenvector represents the axis of the curvature. As per the criteria suggested by Suresh, Gadre and co-workers to characterize a lone pair,⁴¹ the largest eigenvalue and the corresponding eigenvector of the Hessian matrix at $V_{\min}$ are found to be in the direction of N-centers (Fig. S1, ESI†) and the largest eigenvalue (0.0926) shows a significantly larger value than the other two eigenvalues (0.0245 and 0.0225). The topographical features of Fig. 2c and d represent the MESP $V_{\min}$ features of the 5-NO₂-phen and 5-NH₂-phen ligands, respectively. Compared to phen, the change in $V_{\min}$ ($\Delta V_{\min}$) observed for 5-NO₂-phen is 9.04 kcal mol⁻¹ and that for 5-NH₂-phen is -2.95 kcal mol⁻¹ indicating the strong electron withdrawing effect of NO₂ and electron donating influence of NH₂ substituents, respectively. We also computed the MESP at both N nuclei (V_N) as it shows a significant change with respect to the nature of the substituent on the aryl ring. For instance, V_N of the phen ligand (-18.4095 a.u.) is changed by 10.72 kcal mol⁻¹ for 5-NO₂-phen and -2.51 kcal mol⁻¹ for 5-NH₂-phen. The quantities $\Delta V_{\min}$ and ΔV_N (change in MESP at N nucleus) are used as a measure of the substituent effect.^{42–44}

(c) MESP analysis and interaction energy of [Mo(CO)₄(phen*)] complexes

The MESP isosurface plots given in Fig. 3a for Mo(CO)₄ show the lone pair regions of carbonyl groups. The two *trans* disposed carbonyl groups show a significantly higher MESP minimum value, -13.2 kcal mol⁻¹ ($V_{\min-\text{CO}}$), than the other two carbonyl groups, -24.0 kcal mol⁻¹, due to the *trans* influence.^{45,46} As

shown in Fig. 3b, coordination of the phen* ligand to Mo describes the donation of two lone pairs from the nitrogen centers. This donation increases the electron rich character of the carbonyl groups in [Mo(CO)₄(phen)] compared to those in Mo(CO)₄ and accordingly $V_{\min-\text{CO}}$ of [Mo(CO)₄(phen)] becomes more negative by 14.8 kcal mol⁻¹ for the *trans* disposed carbonyl groups and by 13.4 kcal mol⁻¹ for the other two carbonyl groups. Compared to [Mo(CO)₄(phen)], $V_{\min-\text{CO}}$ of [Mo(CO)₄(5-NO₂-phen)] becomes less negative by 4.7 kcal mol⁻¹ for the *trans* carbonyl groups and by 4.9 kcal mol⁻¹ for the other two carbonyls whereas in the case of [Mo(CO)₄(5-NH₂-phen)], $V_{\min-\text{CO}}$ becomes more negative by 1.0 kcal mol⁻¹ for the *trans* carbonyl groups and by 1.4 kcal mol⁻¹ for the other two carbonyls. The $V_{\min-\text{CO}}$ data are in accordance with the electron withdrawing nature of the NO₂ substituent and the electron donating nature of the NH₂ substituent.

In Table 1, $\Delta V_{\min}$ and ΔV_N of all phen* ligands and $\Delta V_{\min-\text{CO}}$ of all [Mo(CO)₄(phen*)] complexes are presented along with the experimental CO stretching frequency $A_{1\text{ax}}$ and the calculated interaction energy (E_{int}) for the complex formation between the phen* ligand and Mo(CO)₄. The calculated interaction energy is found to agree very well with the interaction energy reported by Brenna *et al.* using the ETS-NOCV^{47–50} approach at the PBE0 level (mean absolute deviation is 0.85 kcal mol⁻¹). All the three MESP parameters, *viz.* $\Delta V_{\min}$, ΔV_N and $\Delta V_{\min-\text{CO}}$, show a strong linear correlation with $A_{1\text{ax}}$ with a correlation coefficient (r) of 0.995 (Fig. 4a), 0.996 and 0.967, respectively. Similarly, $\Delta V_{\min}$,

Table 1 MESP properties ΔV_N , $\Delta V_{\min}$ and $V_{\min-\text{CO}}$, $A_{1\text{ax}}$ values and calculated E_{int} values. All MESP values and E_{int} are in kcal mol⁻¹ and $A_{1\text{ax}}$ are in cm⁻¹

S. no.	Name	ΔV_N	$\Delta V_{\min}$	$V_{\min-\text{CO}}$	$A_{1\text{ax}}$	E_{int}
1	phen	0	0	-36.8	2015.1	-58.9
2	5-NO ₂ -phen	10.7	9	-31.9	2017.4	-56.2
3	5-NH ₂ -phen	-2.5	-2.9	-38.3	2014.1	-59.7
4	Bathophen	-2.9	-2.5	-38.8	2013.9	-60.2
5	4,7-Me ₂ -phen	-4.8	-3.8	-39.1	2013.7	-60.2
6	tmphen	-7.8	-5.5	-40.3	2013.1	-61.0
7	5,6-Me ₂ -phen	-3.1	-3	-38.2	2014.2	-59.9
8	5-Me-phen	-1.4	-1	-37.5	2014.7	-59.3
9	5-Cl-phen	4.4	3.8	-35.1	2016.3	-57.7
10	5-Br-phen	4.5	4.1	-35.1	2016.2	-57.7
11	4,7-Cl ₂ -phen	7.8	6.7	-33.6	2017.0	-56.9
12	3,8-Br ₂ -phen	12.1	11.3	-33.8	2018.0	-55.0
13	Br ₄ -phen	15.9	14.4	-32.3	2019.0	-52.6

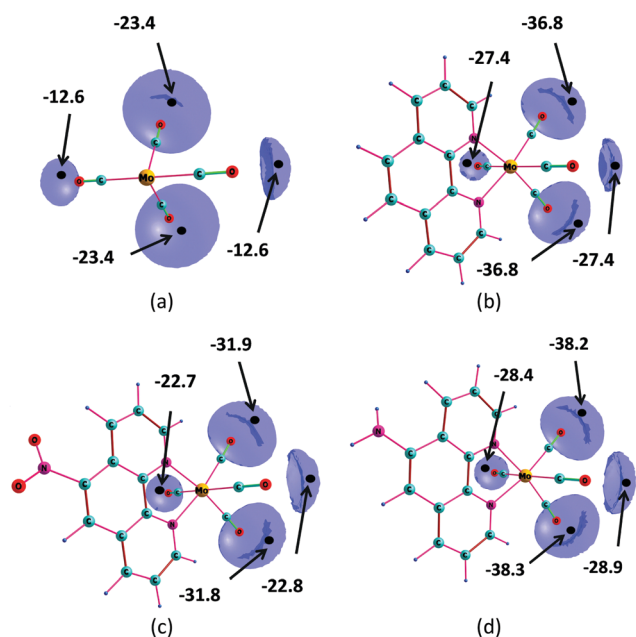


Fig. 3 MESP plots of $V_{\min}$ on carbonyls are shown with (a) 9 kcal mol⁻¹ isosurfaces, of Mo(CO)₄, (b) 25 kcal mol⁻¹ isosurfaces, of [Mo(CO)₄(phen)], (c) 19 kcal mol⁻¹ isosurfaces, of [Mo(CO)₄(5-NO₂-phen)], and (d) 25 kcal mol⁻¹ isosurfaces, of [Mo(CO)₄(5-NH₂-phen)]. $V_{\min}$ values are given in kcal mol⁻¹.

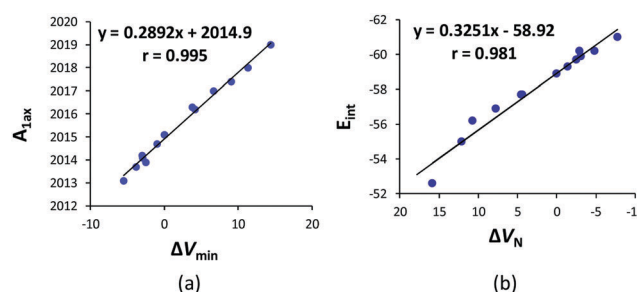


Fig. 4 Correlation of (a) $\Delta V_{\min}$ (kcal mol⁻¹) against $A_{1\text{ax}}$ frequency (cm⁻¹) and (b) ΔV_N (kcal mol⁻¹) against E_{int} (kcal mol⁻¹).

ΔV_N and $\Delta V_{\text{min-CO}}$ exhibit a linear correlation with E_{int} and the corresponding r values are 0.986, 0.981 (eqn (5) and Fig. 4b) and 0.918. These correlations indicate that the electron rich character of N centers measured in terms of the MESP parameter V_{min} or V_N has the power to predict the strength of a phen* ligand to coordinate with a metal center decorated by other ligands. Further, the outcome of the coordination can also be measured in terms of the change in MESP observed at the sensitive ligands in the complex. Here carbonyl groups show a significant change in the MESP value due to electron donation from the phen* ligand to metal and that change is also proportional to the Tolman electronic parameter, the CO stretching frequency.^{51–53}

$$E_{\text{int}} = 0.3251(\Delta V_N) - 58.92 \quad (5)$$

(d) Correlation between the interaction energy and the MESP parameter of benzene derivatives

Previously, a change in the MESP value at the para carbon of a monosubstituted benzene relative to benzene (ΔV_C) has been proposed as an electronic descriptor to quantify the substituent effect which showed a strong correlation with the well-known Hammett σ_p constant.¹⁸ Further, on the basis of the magnitude of ΔV_C , a large variety of 381 substituents have been classified into various categories, viz. very strong ($S^{\gg}$), strong (S^{strong}), medium (S^{medium}) and weak (S^{weak}) electron donating and very strong ($S^{\ll}$), strong (S^{strong}), medium (S^{medium}) and weak (S^{weak}) electron withdrawing. The ΔV_C based assessment of the substituent effect is found to be transferable and additive for conjugated systems such as pyridine, pyrene, butadiene and butadiyne. If phenanthroline systems obey the transferability property of the substituent effect, a strong similarity in ΔV_C and ΔV_N parameters can arise which is true as a strong linear correlation exists between them (Fig. 5). As per the correlation plot in Fig. 5 for the 13 substituents,

$$\Delta V_N = 0.5519(\Delta V_C) \quad (6)$$

Substituting ΔV_N in eqn (5) leads to

$$E_{\text{int}} = 0.1794(\Delta V_C) - 58.92 \quad (7)$$

ΔV_N is approximately 0.5519 times ΔV_C which indicates that without doing any calculation on phenanthroline derivatives,

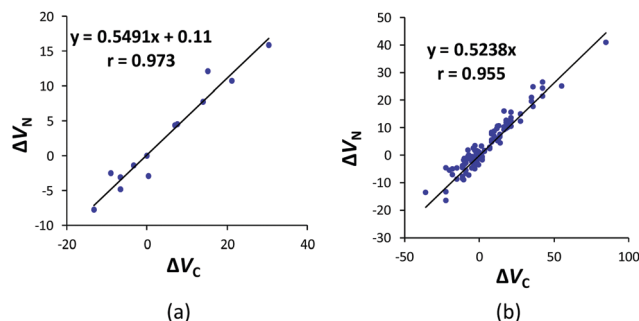


Fig. 5 Correlation between ΔV_C and ΔV_N values for (a) the thirteen phenanthrolines and (b) the 108 ligands given in Table 2. All values are in kcal mol^{−1}.

Table 2 The predicted E_{int} using eqn (5) ($E_{\text{int-eq5}}$) and actual ΔV_N values for 46 systems. Values are in kcal mol^{−1}

Ligand	ΔV_N	$E_{\text{int-eq5}}$	Ligand	ΔV_N	$E_{\text{int-eq5}}$
2-NO ₂	15.7	−53.8	3,4-(OH) ₂	−1.6	−59.4
2-CN	12.7	−54.8	3,4-(OMe) ₂	−3.4	−60
2-CHO	6.7	−56.7	4,7-(NO ₂) ₂	24.3	−51
2-OCF ₃	8.4	−56.2	4,7-(OCF ₃) ₂	9.3	−55.9
2-CH=CH ₂	−0.7	−59.1	4,7-(CH=CH ₂) ₂	−1.5	−59.4
3-NO ₂	13.7	−54.5	3,8-(NO ₂) ₂	26.5	−50.3
3-CN	11.9	−55	3,8-(CHO) ₂	15.1	−54
3-CHO	7.6	−56.5	3,8-(SiH ₃) ₂	2.9	−58
3-OCF ₃	7.8	−56.4	5,6-(CN) ₂	17.8	−53.1
3-CH=CH ₂	0.6	−58.7	5,6-(OCF ₃) ₂	10.3	−55.6
4-NO ₂	12.5	−54.9	5,6-(CH=CH ₂) ₂	−0.6	−59.1
4-CN	11	−55.3	3,4-7,8-(NO ₂) ₄	41	−45.6
4-CHO	7.4	−56.5	3,4-7,8-(CHO) ₄	25.1	−50.8
4-OCF ₃	4.8	−57.3	3,4-7,8-(SiH ₃) ₄	4.5	−57.5
4-CH=CH ₂	−1.1	−59.3	2,9-(CN) ₂	24.8	−50.9
5-NO ₂	10.6	−55.5	2,9-(OCF ₃) ₂	16.1	−53.7
5-CN	9.8	−55.7	2,9-(CH=CH ₂) ₂	−1.6	−59.5
5-CHO	6.8	−56.7	4,7-NO ₂ ,CHO	21	−52.1
5-OCF ₃	5.6	−57.1	4,7-CN,CH=CH ₂	11.9	−55
5-CH=CH ₂	0.5	−58.8	4,7-OCF ₃ ,SiH ₃	9.7	−55.8
3,4-(NO ₂) ₂	21.5	−51.9	3,8-NO ₂ ,CHO	19.7	−52.5
3,4-(CHO) ₂	12.4	−54.9	3,8-CN,CH=CH ₂	10.2	−55.6
3,4-(SiH ₃) ₂	2.4	−58.1	3,8-OCF ₃ ,SiH ₃	6.1	−56.9

the knowledge of ΔV_C derived from simple benzene derivatives alone can be utilized for predicting the strength of the complex formation between phen* and Mo(CO)₄ via eqn (7). In order to test this hypothesis, representative sets of substituents from various categories, viz. $S^{\gg}$, S^{strong} , S^{medium} , S^{weak} , $S^{\ll}$, S^{strong} , S^{medium} and S^{weak} , are selected and populated at different positions of the phenanthroline ligand to create 108 substituted phenanthroline ligands which contain mono-, di- and tetra-substituents.¹⁸ For di- and tetra-substituted systems, ΔV_C values were added twice and four times, respectively, assuming the additive nature of the substituent effect.¹⁸

The predicted E_{int} using eqn (5) ($E_{\text{int-eq5}}$) and the actual ΔV_N values of 46 systems are presented in Table 2 while the values of the remaining 62 systems are given in Table S2 in the ESI.† Also the predicted E_{int} using eqn (7) ($E_{\text{int-eq7}}$) and predicted ΔV_N using eqn (6) ($\Delta V_{N\text{-eq6}}$) and ΔV_C values are presented in Table S3 in the ESI.† The mean absolute deviation between the actual ΔV_N and $\Delta V_{N\text{-eq6}}$ is 2.23 kcal mol^{−1} while that between $E_{\text{int-eq5}}$ and $E_{\text{int-eq7}}$ is 0.72 kcal mol^{−1} indicating that the quantification of the substituent effect empowers us to make strong predictions on the binding affinity of a large variety of substituted phenanthroline derivatives to Mo(CO)₄. The additivity and transferability properties of the substituent effects are also well evident in the good agreement seen in the ΔV_N and $\Delta V_{N\text{-eq6}}$ based predictions on interaction energy.

(e) Direct prediction of interaction energy solely by the substituent effect parameter

The MESP based analysis of interaction energy clearly suggests that the coordination behavior of a phenanthroline ligand to a metal complex is very much predictable provided the substituent effect is known for them. The MESP based parameters ΔV_C and ΔV_N emerge as effective measures of the substituent effect.

Using the ΔV_C values from the database, *a priori* prediction on the coordination behavior of phenanthroline ligands to metal complexes can be achieved without doing a calculation on the ligand or the complex whereas more precise predictions can be obtained by calculating ΔV_N for each ligand. In order to rigorously test these hypotheses, a set of 50 substituents are considered along with their ΔV_C values from our substituent database.¹⁸ The E_{int} is predicted for mono-, di- and tetra-substituted systems using eqn (8) which is a generalized version of eqn (7) for a multiply substituted phen*.

$$E_{\text{int}} = 0.1794(\sum(\Delta V_C)) - 58.92 \quad (8)$$

Here, assuming additivity⁵⁴ of substituent effects, $\sum(\Delta V_C)$ represents the total ΔV_C values due to all the substituents. Table 3 gives the multiple effect of the same substituent on the phen* ligand (E_{int} of 25 systems are given and the remaining 25 values are given in Table S4 of the ESI†). In order to analyze the multiple effect of different substituents, we have generated 5000 tetra-substituted phen* ligands by randomly populating four different substituents from categories such as $S^{\gg}$, $S^{\gg\gg}$, $S^{\gg\gg\gg}$, $S^{\gg\gg\gg\gg}$, $S^{\ll}$, $S^{\ll\ll}$, $S^{\ll\ll\ll}$ and $S^{\ll\ll\ll\ll}$ at various positions (ESI†) on the aromatic ring. Fig. 6 shows the relationship between ΔV_C and E_{int} for all the 5000 randomly generated ligands. The predicted E_{int} values of a representative set of tetra substituted derivatives are provided in the ESI† (Table S3). Also the depiction of the E_{int} of the 13 experimentally known systems along the prediction line (Fig. 6) clearly reflects the use of ΔV_C as an effective parameter to quantify the substituent effect.

It may be noted that ΔV_C based prediction of E_{int} for a multiply substituted system does not take into account the influence of the position of the substituent in the phenanthroline ligand as the

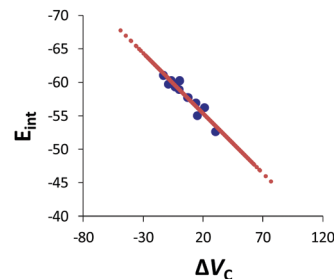


Fig. 6 Correlation between ΔV_C and E_{int} for 5000 randomly generated combinations. The blue dots stand for the experimentally known systems. All values are in kcal mol⁻¹.

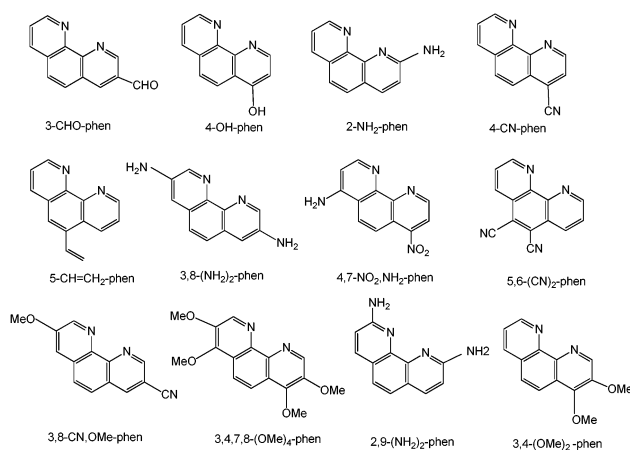


Fig. 7 Twelve phenanthroline derivatives for comparing the predicted E_{int} with calculated E_{int} values.

Table 3 ΔV_C values and $E_{\text{int-eq8}}$ values for mono, di and tetra combinations of 25 selected substituted phen* ligands. All values are in kcal mol⁻¹

S. no.	Substituent	ΔV_C	$E_{\text{int}}(\text{mono})$	$E_{\text{int}}(\text{di})$	$E_{\text{int}}(\text{tetra})$
1	COCl	19.1	-55.5	-52.1	-45.2
2	CN	18	-55.7	-52.5	-46
3	SO ₂ Me	16.8	-55.9	-52.9	-46.9
4	Cyclo-C ₄ F ₇	16.1	-56	-53.1	-47.4
5	C(Et)(NO ₂) ₂	15.4	-56.2	-53.4	-47.9
6	CF ₂ CF ₃	14.7	-56.3	-53.6	-48.4
7	OCN	14.2	-56.4	-53.8	-48.7
8	CHO	13.8	-56.4	-54	-49
9	ONO ₂	13.4	-56.5	-54.1	-49.3
10	OCOCF ₃	12.2	-56.7	-54.5	-50.2
11	CONHMe	6.4	-57.8	-56.6	-54.3
12	C≡CH	5.7	-57.9	-56.9	-54.8
13	SiClMe ₂	4.8	-58.1	-57.2	-55.5
14	SCH=CH ₂	4.1	-58.2	-57.4	-56
15	C ₆ H ₄ -3Cl	3.6	-58.3	-57.6	-56.3
16	NHCONH ₂	-3.7	-59.6	-60.2	-61.6
17	Cyclohexyl	-3.8	-59.6	-60.3	-61.6
18	Et	-4	-59.6	-60.4	-61.8
19	NHC ₆ H ₅	-4.8	-59.8	-60.6	-62.4
20	OMe	-5	-59.8	-60.7	-62.5
21	O(CH ₂) ₃ CH ₃	-5.9	-60	-61	-63.2
22	OCHMe ₂	-6.5	-60.1	-61.3	-63.6
23	NH ₂	-9	-60.5	-62.1	-65.4
24	NHNH ₂	-11.2	-60.9	-62.9	-67
25	NMe ₂	-12.3	-61.1	-63.3	-67.7

model is based on the additive nature of the substituent effect. For instance, such a prediction may deviate from the actual value if the electronic effects exerted by the substituent at different positions are significantly different or significant steric influence arises due to the occupancy of substituents at adjacent positions. To verify this feature, the E_{int} values of 12 phenanthroline derivatives given in Fig. 7 have been calculated and compared with the predicted E_{int} using eqn (5) and (7) and the data are depicted in Table 4. In general, majority of the systems obey the additivity rule to a great

Table 4 Calculated E_{int} values of 12 representative phenanthroline derivatives along with the E_{int} values predicted using eqn (5) and (7). Values are in kcal mol⁻¹

S. no.	Complex	E_{int}	$E_{\text{int-eq4}}$	$E_{\text{int-eq2}}$
1	[Mo(CO) ₄ (3-CHO-phen)]	-56.4	-56.4	-56.5
2	[Mo(CO) ₄ (4-OH-phen)]	-59.2	-59.3	-59.8
3	[Mo(CO) ₄ (5-CH=CH ₂ -phen)]	-61.0	-58.7	-58.8
4	[Mo(CO) ₄ (2-NH ₂ -phen)]	-59.1	-60.5	-60.7
5	[Mo(CO) ₄ (4-CN-phen)]	-57.3	-55.7	-55.3
6	[Mo(CO) ₄ (5,6-(CN) ₂ -phen)]	-54.5	-52.5	-53.1
7	[Mo(CO) ₄ (4,7-NO ₂ ,NH ₂ -phen)]	-55.3	-56.7	-55.5
8	[Mo(CO) ₄ (3,8-CN,OMe-phen)]	-54.4	-56.6	-56.9
9	[Mo(CO) ₄ (3,8-(NH ₂) ₂ -phen)]	-60.5	-62.1	-60.5
10	[Mo(CO) ₄ (3,4-7,8-(OMe) ₄ -phen)]	-55.1	-62.5	-60.7
11	[Mo(CO) ₄ (2,9-(NH ₂) ₂ -phen)]	-58.0	-62.2	-62.3
12	[Mo(CO) ₄ (3,4-(OMe) ₂ -phen)]	-54.1	-60.7	-60.0

extent while significant deviation occurs for systems showing the steric effect such as 3,4-7,8-(OMe)₄-phen and 3,4-(OMe)₂-phen.

Conclusions

Quantum-chemical methods and molecular modelling techniques enable the accurate description of a large number of molecular properties and their effective utilization leads to the understanding of primitive patterns of molecular reactivity.^{55,56} Here the MESP, an electronic property of the molecular system, is utilized to follow the electron rich/deficient character of the lone pair bearing nitrogen centers in substituted 1,10-phenanthroline systems. The primitive pattern of molecular reactivity is evident in the MESP as it identifies a more electron rich nitrogen center from an electron deficient one and interprets the higher binding affinity of the former with Mo(CO)₄ solely based on the MESP parameter V_N . The electron donation from the phenanthroline ligands to the metal center can be correlated with the increase in the negative character of MESP observed at CO ligands or to the decrease in CO stretching frequency. The MESP based quantification and classification of substituent effects deduced for simple benzene systems are found to be applicable for the 1,10-phenanthroline ligands which indicates that substituent effects obey the transferability property. The binding energy between the ligand and Mo(CO)₄ is highly predictable from the knowledge of the total substituent effect experienced by the ligands which supports the additivity property of substituent effects. The MESP analysis is also shown as a general approach to fine tune the electronic properties of 1,10-phenanthroline ligands to achieve the highest coordination power of the ligand towards a transition metal center.

Conflicts of interest

There are no conflicts to declare.

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